On the “Plasmaticness” of Animation in the Context of AIGC

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Film Art, 2025, No. 3 (Issue 422), pp. 88-95

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Abstract

As artificial intelligence generation technology emerges as a new kind of digital object in the contemporary era and a new form through which animation art can be understood, an examination of its technical core and visual characteristics reveals unpredictable, fluid transformations. These conceal a continuously emerging, cell-like force of growth and a dissolution of the human author's subjectivity. From animation's beginnings to the present, its images have always carried unstable transformations akin to “protoplasm.” Examining the threefold transformation of contour, form, and subject reveals a vitality that persists throughout changes in animation's media. Exploring the continuities and evolution of its “plasmaticness” may thus enable an ontological account of AIGC imagery as animation.

Keywords: Sergei Eisenstein; plasmaticness; AIGC animation; ontology of animation

Introduction

Since 2023, the emergence of large models for artificial intelligence-generated content (AIGC) has propelled artistic creation into a new era. In artistic creation and media production, AIGC refers both to content generated autonomously or with assistance through the algorithms and technologies of large AI models, and to the process of generating such content. The AIGC White Paper (2022), released by the China Academy of Information and Communications Technology and the JD Explore Academy, identifies the corresponding international term as “AI-generated media,” defining it as “a general term for the production, manipulation, and modification of data or media through artificial intelligence algorithms.”[1]

AIGC technology has undergone multimodal transformation and iteration. It began with textual instructions and dialogue, represented by chatbots such as ChatGPT, followed by large image models such as Midjourney, which generate images from pictures or text prompts. Now, after successive rounds of model development and refinement, Sora, a large video model from OpenAI, the company behind GPT, has emerged, bringing public attention to and discussion of AIGC imagery to unprecedented heights.

Whether animation films are generated frame by frame with image models such as Stable Diffusion, or produced by video models such as Sora through text-to-video, image-to-video, or video-extension instructions, their outward characteristics include deformation, fluidity, changes of scale, and flickering. In their formal features and visual effects, they exhibit a “plasmaticness” similar to that of early traditional two-dimensional animation.[2] This concept originated with the Soviet film director and theorist Sergei Eisenstein. Deeply inspired by his visit to the Disney studio in the spring of 1930, he first proposed the theory of plasmaticness. As a biological term, protoplasm denotes the fundamental substance constituting cells, principally their active internal material, which directly participates in life processes. Cells maintain vital functions through the biochemical reactions within protoplasm, including metabolism, division, and responses to stimuli. Drawing on this biological metaphor, Eisenstein termed the bodily deformation and contraction he observed in Disney characters “plasmaticness”: Mickey Mouse's hands can extend far

beyond their ordinary length, while his body can expand, contract, and disintegrate, assuming endlessly varied forms. The concept here means that characters, like primordial protoplasm, lack a stable form but can assume "any form" and "leap along the evolutionary ladder."[3] For Eisenstein, plasmaticness symbolizes a primordial material whose liquid state contains unlimited potential, embodying animation's freedom of formal transformation. Animation is thereby endowed with a revolutionary power to resist standardized capitalist production, and plasmaticness reveals the medium's potential.

In this sense, Eisenstein's theory of plasmaticness has been regarded as an ontological characteristic of animation. Yet as animation developed, especially with the rise of CGI, traditional realist simulation and conceptions of realism paradoxically dissolved animation's plasmaticness to some extent. Nevertheless, the emergence of AIGC imaging technologies has opened entirely new dimensions of creation. The resulting questions are these: Can AI-generated imagery be regarded as animation? Within Eisenstein's theoretical framework, can animation's plasmaticness still release its cultural potential?

The Development and Evolution of Animation's "Plasmaticness"

Although Eisenstein's theory was inspired by Disney animation, plasmaticness as an artistic characteristic has manifested itself throughout animation's history. It has evolved alongside advances in animation technologies and media, exhibiting different media potentials and narrative characteristics. In AIGC animation in particular, this plasmaticness reappears and further expands narrative possibilities and visual innovation, marking another dynamic transformation and evolution of animation within its technological and cultural contexts.

As early as the late eighteenth and early nineteenth centuries, images produced through optical technologies and visual experiments already exhibited animation's basic characteristics and can even be regarded as a form of "pre-animation."[4] Early optical devices, including moving-image magic lanterns, optical theaters, and other apparatuses, produced continuously transforming and reshaping images because of the technical limitations of their optical principles. These images broke free of real-world physical constraints and displayed a distinctive fluidity and plasticity. In Ars Magna Lucis et Umbrae, Athanasius Kircher described his "metamorphosis machine" and "phantasmagoria" in detail. In Kircher's "mirror box," light underwent multiple refractions, projecting static images attached to the apparatus onto a wall along a conical projection path. Through this optical device, viewers could see their own image continuously transformed into composite images with the head of a donkey, deer, or eagle. In this environment of light and shadow, human figures "appeared and disappeared, as if coming and going," and "grew and shrank of their own accord, apparently freed from technical and material support."[5] Such unpredictable distortion and transformation directly express the plasmaticness of early animation.

Eisenstein's central account of plasmaticness derived primarily from his study of Disney's hand-drawn animation of the 1920s and 1930s. In Eisenstein on Disney, he examined films including Mickey Mouse (1928) and The Skeleton Dance (1929), identifying the characters' plasmaticness as the source of their initial appeal. He observed that in Steamboat Willie (1928), Mickey Mouse's arms can stretch "far beyond the length allotted to them" (Figure 1), while his body can continually expand, contract, and disintegrate. For example, Mickey removes the upper part of his head to greet Minnie (Figure 2). Characters can turn into different objects at any moment: a monkey's neck becomes a spring, an octopus becomes an elephant, and a striped fish becomes a tiger. For Eisenstein, Mickey Mouse is a creature "behaving like primordial protoplasm," which has not yet acquired a "'stable' form" but can assume any form, leap along the evolutionary ladder, and attach itself to any animal form.[6]



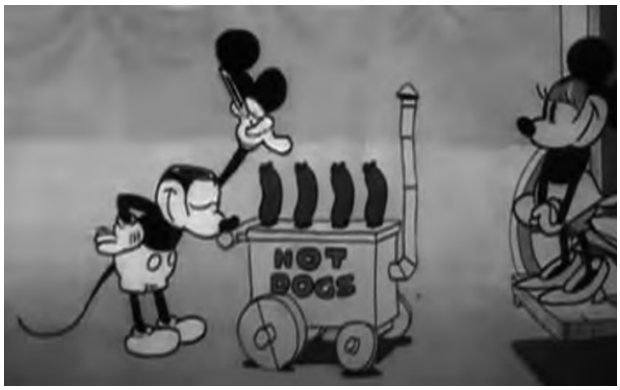


Figure 1. Mickey Mouse's bodily transformation in Steamboat Willie (left). Figure 2. Mickey Mouse doffing his hat in The Karnival Kid (right).

Beyond analyzing visual form, Eisenstein's discussion of Disney and plasmaticness also sought to bridge spectators' experience and social reality, making animation a link between established ideology and historical materialism. He argued that Disney animation's capacity to connect forms without spatial or temporal constraints gives viewers more than a "moment of oblivion" from Taylorist production:[7] it also dismantles traditional classificatory systems. In Merbabies (1938), for example, a caged striped fish abruptly becomes a tiger and emits a cross-species roar, while an octopus's tentacles turn into an elephant's trunk. Such transformations that "depart from the self"[8] challenge conventional rules of classification. In this context, plasmaticness transgresses biological boundaries and destroys the cognitive frameworks attached to existence, from norms of animal behavior and prescribed social roles to the certainty of spatial and temporal coordinates. All are dismantled through animation. When animation permits creatures to "leap along the evolutionary ladder,"[9] it suspends the differentiating systems on which modernity depends, reducing existence to a field of pure possibility and resisting the established order of capitalist life.

Table 1. Stages in the development and evolution of animation's plasmaticness.

Stage	Animation type	Aesthetic feature	Audiovisual characteristics
Emergence	Optical experimental animation	"Transformation" and "plasticity"	Unpredictable, marvelous distortions and transformations of light and shadow produced through optical principles.
Flourishing	Disney hand-drawn animation	"Plasmaticness"	Characters lack a "stable" form but can assume any form.
Decline	CGI animation	"Second-order realism"	Highly realistic simulation suppresses exaggerated deformation and the possibility of free expression.
Return	AIGC animation	"Dynamic growth"	Technological materiality constructs a decentered, de-subjectified "plasmatic body," expressed through "threefold transformation."

Since Pixar's Toy Story (1995), the first fully CGI animated feature in history, CGI animation has gradually become the dominant production method in modern animation. This marks a change in production tools and aesthetic ideas: from the exaggerated stylization of hand-drawn animation toward an aesthetic closer to what Andrew Darley calls "second-order realism."[10] Second-order realism does not seek direct imitation of the real world. Instead, it borrows realist techniques from traditional cinema to

construct within the computer a seemingly realistic world that is fundamentally virtual and hyperreal.

Within this framework, however, extreme verisimilitude suppresses animation's possibilities for exaggerated deformation and free expression, eroding the plasmaticness that animation ought to possess. Characters in many CGI films, such as Zootopia (2016) and Frozen (2013), maintain highly consistent traits, including fixed vocal qualities, skin colors, whiskers, and other details, with few qualitative changes of form. This also diminishes, to some degree, the cultural potential of animation's plasmaticness.



Figure 3. Zootopia.

The rapid development of AIGC technology challenges the realist conception of CGI animation that dominates the screen. In terms of technical logic, the algorithms, instructions, and data underlying AIGC animation construct, through their technological materiality, a decentered, nonhierarchical network of relations and modes of existence without a sovereign subject. It becomes a “plasmatic body” characterized by continuous growth and fluidity. Formally, the transformations in AIGC imagery's audiovisual language allow it to express complex social issues such as time, gender, and class more flexibly, and demonstrate new potential for resisting conventional taxonomic orders. In this sense, AIGC animation undoubtedly marks the revival and evolution of plasmaticness in animation.

From the experiments with light and shadow of pre-animation to hand drawing and the digital production processes of CGI and AIGC, animation has undergone changes in its media functions and aesthetic language. Its plasmaticness has consistently featured free transformation and unstable fluidity, appearing to different degrees at each stage. As a new form of digital production, AIGC imagery reconstructs the existential relationship between human beings and the computer as an object. Its emergence reflects a reshaping of the relationship between technical objects, people, and their arts and culture. The philosopher of technology Yuk Hui proposes the concept of the “digital object”: a relational mode of being that is neither a substance nor merely information. It is “a new kind of industrial product, permeating every aspect of life in our ubiquitous media age - online videos, images, texts ... They exist both on the screen, where we can interact with them, and at the back end of, or inside, computer programs.”[11] As a contemporary digital object, AIGC creates stable and complete associated milieus for human beings, sustaining emotions, atmospheres, collectives, and memory. In this sense, AIGC imagery as a digital object becomes ever more closely related to human beings as technological operators. Digital objects become a crucial means of developing technological and social culture, possessing unlimited cultural potential while continuing the power of plasmaticness embedded in animation.

Technical Logic: “Growth” and “De-subjectification” in AIGC Imagery

Early animation scholars explored the concept of “life” in depth in works such as The Illusion of Life[12] and Eisenstein on Disney. They argued that animation differs from other media precisely through its ability to impart life. Eisenstein in particular explicitly stated that “the very idea of the animated cartoon is like a direct embodiment of the method of animism.”[13] Within this framework, animation is a medium that creates its distinctive artistic existence by giving life.

In discussing how a sense of life emerges in animation, traditional scholars generally begin with audiovisual language. Following the primacy of “formal cause” within Aristotle's four causes, they focus on structure and form, arguing that specific principles of deformation, such as stretch, squash, and anticipation, simulate movement and vitality in animation. By contrast, the process through which life is perceived in AIGC animation returns to a logic in which Aristotle's “material cause” takes priority. The “absence of the human” in AIGC animation's generation allows animation to display the potential for self-development. This opens a new way of giving life, manifesting a capacity for growth and revealing a new technological paradigm for bringing animation to life.

In terms of model algorithms and technical logic, AI animation relies on autoregressive and diffusion models. Having learned from large quantities of textual and image data, these models respond to instructions by generating content that moves from random noise to ordered images. In terms of image generation and media characteristics, AIGC animation displays a capacity for growth that corresponds to certain traits of traditional animation's plasmaticness. Biological growth involves cells combining to constitute a living organism. Through deep learning and self-optimization, large AIGC image models continually generate new content, much as plants absorb nutrients and grow. Algorithms and data play the role of “cells,” giving AIGC imagery a capacity for growth resembling organic life.

In Film Form, Eisenstein writes that “shots should connect with one another like the cells of an organism, jointly advancing the film's internal development.”[14] He also states that “a film should grow naturally, like an organic living being, rather than be deliberately pieced together.”[15] The algorithmic process of “autoregression” exhibits a similar capacity for organic self-growth and self-shaping. In Unconscious Memory, Samuel Butler asserts that “the distinction between the organic and the inorganic is arbitrary; what we call the inorganic world must be regarded as alive to some degree.”[16] AIGC animation technically realizes this expression of growth, presenting a biologically analogous process of self-growth and regulation through an autoregressive mechanism.

The generation of AIGC animation is essentially a probability-based process of dynamic evolution. In particular, the introduction of reinforcement learning gives it an “autoregressive mechanism.” A feature of this model is that, as animation is generated frame by frame, each frame introduces some randomness and fluctuation. Sora, for example, supports multimodal transformations including text-to-video, image-to-video, and video-to-video, and uses historical information in the data to predict the next point in a sequence. Its autoregressive mechanism is applied to video-frame generation, allowing the model to predict the next frame from previously generated frames and textual descriptions. This progressive mode of generation requires algorithms both to infer and calculate from previous states and to adapt to new inputs or adjust strategies through continuous optimization. Animation generation thus resembles an organism's growth, continually adjusting to changes in its environment. In this sense, the animation medium changes from a receptive form passively altered by its creator into an endogenous plasmatic entity capable of self-evolution and growth. The medium is no longer an inert carrier awaiting external instructions but an adaptive “plasmatic body” able to generate, reconstruct, mutate, and flow autonomously.

From a microbiological perspective, protoplasm constitutes the fundamental unit of life, and its organization into communities embodies symbiosis. Interactions among protoplasmic cells resemble the operation of Bruno Latour's "actor-network,"[17] in which parallel actions among multiple agents sustain life. In this framework, protoplasm, as a nonhuman "actor," activates or inhibits particular reactions by exchanging matter and information, constructing "channels" that shape the state of life. This decentralized, dynamic interaction exhibits complex symbiotic characteristics.

In traditional animation, whether frame-by-frame or CGI, the author always occupies the center of power, and production involves precise semiotic-narrative translation. Animation, as a semiotic text addressing philosophical questions and sociocultural issues, consequently becomes an important medium through which researchers explore theories such as posthumanism and semiotics. Researchers commonly analyze, for example, how animation mediates social and philosophical questions, and how authors express ideas through it.

AIGC animation's technical principles mean that generation is not simply an input-output process. It depends instead on AI's statistical analysis of enormous datasets to integrate visual and textual information. Contemporary datasets are vast: the LION 5B dataset, for example, contains 5.8 billion image-text data points.[18] In this environment, textual prompts cannot easily correspond directly to the stable sign relations of semiotics.

Consequently, animation authors' capacity for precise translation in the semiotic sense is weakened, and their control over a work is partly dissolved within technological architectures, data training, and hardware. As the French philosopher Bernard Stiegler writes, "the 'who' of subjectivity begins to be 'objectively deconstructed.'"[19] To some extent, this process echoes Roland Barthes's proclamation of the "death of the author." In discussions of the AIGC filmmaking section of the Tribeca Film Festival, creators spoke of frequently feeling a "loss of control" when using AI animation tools. Claire Evans, for instance, regards AI as a "dynamic nonlinear system" affected by unpredictable factors, making its direction of generation difficult to control.[20]

Yet this loss of control should not be understood solely as a technical limitation; it can also be viewed as an expression of the AIGC medium's dynamic characteristics. The interaction of algorithms, hardware, and data allows AIGC animation to transgress the discursive power of the traditional author. Trillo observes that the failures arising from this loss of control are in fact "ironically positive failures," comparing the process to iterating oneself within a space of possibilities and discovering an unknown self.[21] This de-subjectified discourse of animation is forming a "statement" in the Foucauldian sense. For Foucault, "the statement has no productive subject; it is anonymous and autonomous, and never operates by reference to an 'I think.'"[22] In AIGC animation, interactions between algorithms and datasets mean that generation no longer depends on a "subject" in the traditional sense. Its boundaries become blurred, and it ceases to be controlled by a single subject. This "death of the author," however, does not simply remove the human creator. It reconfigures roles within creation, enabling democratic interaction and possible symbiosis between "the human" and the animation medium - its algorithms, data, and model training.

Formal Characteristics: "Threefold Transformation" in AIGC Imagery

As discussed above, AIGC's technical capacities for growth and the displacement of authorship jointly accomplish the giving of life through "material cause." From the perspective of "formal cause," meanwhile, transformation constitutes the aesthetic form of plasmaticness.

Transformation was initially widely used as an artistic device in literature. Within animation, Norman Klein,[23] Eisenstein, and others have regarded it as a core ontological characteristic. Eisenstein argues that

Disney inherits literature's "principle of transformation": "In his very design, Mickey truly embodies the 'ideal of the Bororo' - he is both a human being and a mouse!"[24] The art historian Panofsky remarks that "the true merit of animated cartoons lies in making things animate, that is, giving life to what is lifeless ... This produces a metamorphosis, expressed to the fullest in Disney's animals, plants, thunderclouds, and railway trains."[25] Through transformation, animation turns originally static or abstract images into visual expressions full of life and movement. It creates fluid connections between images and narrative transitions through animation itself, rather than relying on editing.

Maria Pags, a professor of animation at the Polytechnic University of Catalonia, categorizes Eisenstein's aesthetic account of plasmaticness in Eisenstein on Disney, identifying three kinds of transformation: "plasmaticness of contour," "plasmaticness of form," and "plasmaticness of subject."[26] In AIGC animation, plasmatic transformation both continues forms of expression found in traditional animation and develops and expands them in certain respects.

In the third chapter of Eisenstein on Disney, "Metamorphosis and Animation," Eisenstein examines transformation in Disney animation, arguing that the movement of lines is the most immediate manifestation of animation's plasmaticness. He traces the history of lines back to the earliest cave paintings, where an outline was merely "a line, only a contour." Such simple delineation was not a conscious creative act but a natural expression through the "wandering eye," reflecting a state in which "the hand had not yet separated itself from this movement as an independent movement."[27] When these static lines enter animation, however, they transcend their static limitations and become dynamic carriers with a vitality of their own.

In animation technique, "plasmaticness of contour" is based on the two-dimensional principle of "squash and stretch." Through it, characters' contours deform according to their interactions and biological characteristics. Eisenstein argues that this deformation gives characters a vivid sense of life. Thomas Lamarre explains that the elastic deformation of contour lines allows viewers to perceive an "inner vitality," constructing animation's sense of life through a dialectic of "deanimation-reactivation." Through this process, animation not only reproduces movement but also gives rise to "suprasubjective" forms of life.[28]

With technological development, contours in AIGC imagery are no longer confined to two-dimensional lines. They are defined instead as the edges of algorithmically generated visual images or objects. Because neural networks draw training data from multiple artistic styles, generated images adjust the dynamics of their lines through "scene adaptation." They transcend the physical limitations of traditional squash and stretch, displaying fluid lines perpetually on the verge of disintegration. In the AIGC animation Capital of Conformity (2023), lines may at a given moment become liquid and undulating, or seem to be engulfed by air, light, and shadow, producing a dynamic and continually changing "awareness of contour" (Figure 4). In the roller-coaster sequence in particular, as visitors' contour lines continually disintegrate, viewers cannot predict the trajectory of transformation and are drawn into what Deleuze describes as a "pure audio-visual situation."[29] Here, "the optical and sound situation no longer results from an action, nor is it extended into action; it is neither an index nor a synsign."[30] The expression of life exceeds traditional suprasubjective boundaries and flows through the image's disintegration. Each rupture of a line challenges established cognitive frameworks, while the subsequent process of reassembly becomes a dynamic field in which subject and image jointly construct meaning.



Figure 4. Changes of contour in Capital of Conformity.

In Maria Pags's theoretical framework, “plasmaticness of form” refers chiefly to transformations of a character's external shape. Such transformations include animals becoming objects, humans becoming animals, and one animal becoming another. Through these changes, animation transcends biological limitations, gives fluidity and a sense of life to objects and creatures, and reshapes their formal characteristics within the animated world.[31] Unlike plasmaticness of subject, this dimension of formal transformation does not involve changes in the character's internal behavior.

Maria Pags's elaboration of plasmaticness of form derives primarily from Eisenstein's discussion in the second chapter of Eisenstein on Disney. Eisenstein notes that the continual changes in characters' forms in Shakespeare's comedies generally involve bodily transformation through disguise or magic. In Disney animation, formal transformation is widely employed as a central strategy. Mickey, for instance, as a humanized animal figure, is effectively a formal transformation of human behavior. Moreover, examples of Disney characters turning into one another abound: in Merbabies, the little mermaids' forms as splashes of water eventually return to solid bodies.

Formal transformation is especially common in AIGC creation because of the technology's high flexibility and nonlinear plasticity, which enable freer and more variable transformations in character design. The Cannes award-winning animated short The Crow (2021) feeds frames from the live-action short Painted (2015) into OpenAI's CLIP neural network and generates images of a dancing crow from the dancer's moving-image data (Figure 5). Because of these dual data inputs, the crow's form remains in continuous transition between “crow” and “human.” Its dance is not a linear, predictable moving image but a continual shifting and reassembly between the two forms, conveying a force of resistance between life and death against a barren, impoverished background. This language of AIGC-based formal transformation is fundamental to its ability to exceed traditional animation's limits. It allows more complex visual expression while opening new dimensions for emotional and philosophical articulation.



Figure 5. The live-action short Painted (left) and the AIGC animation The Crow (right).

The dimension of “transformation as subject” emphasizes fundamental changes in a character's behavior, personality, and subjectivity: as form changes, the character's subjective identity changes too. In the second chapter of Eisenstein on Disney, Eisenstein states that, by transforming characters' forms, Disney animation “returns humanity to those earlier stages described by Darwin.”[32] In Merbabies, a caged striped fish becomes a tiger, roaring like a lion or leopard; an octopus becomes an elephant; and a fish becomes a donkey. These formal changes show characters departing from their “selves,” breaking through the names, forms, and behavioral norms once and forever prescribed for them, and constituting the third dimension of plasmaticness.

Compared with traditional animation, AIGC animation displays a de-subjectified mode of representation. In the context of the “death of the author,” AIGC's sign system loses its traditional stability. Characters no longer possess fixed identities and appearances but become random products generated from statistical probabilities within datasets. This technical characteristic places AIGC characters in a fluid, mutating state, disrupting the established identity structures of traditional animation.

Specifically, when AIGC animation handles human figures, especially faces, it is often affected by the technical bottleneck of insufficient noise stability in diffusion models. Faces, as a central feature of traditional animation and live-action imagery, usually help viewers establish a character's identity. In AIGC animation, however, the use of different datasets often results in faces that lack continuity and consistency, disrupting the subject's unity. Although this technical limitation is a bottleneck in AIGC animation's development, it also becomes a key feature of its distinctive narrative forms. In To Dear Me (2024), which won first prize at the inaugural Venice International Film Festival of AI-generated short films, the girl's face remains in a state of flickering, dynamic change and ceases to possess conventional physiognomic characteristics. This disrupts viewers' traditional understanding of the subject's unity. Through innumerable changing faces, the film can also present the diverse and complex conditions of women's lives. In this sense, the subjectivity of AIGC characters resembles the “nomadic subject” defined by Deleuze and Guattari.[33] A nomadic subject is not a fixed individual with a consistent identity in the traditional sense, but a mode of existence that flows and mutates among multiple forces and fields.

Conclusion

The richness of animation's aesthetic forms and the freedom of its imagination are extended by AIGC technology. Its unlimited potential makes animation a new domain of creative possibility. More than a century ago, Eisenstein discovered the formal characteristics of plasmaticness in early animation's transformations and recognized its powerful vitality as an image-based art. Today, amid the rapid development of new information media in the twenty-first century, AIGC imagery blurs the central position occupied by human subjectivity in traditional artistic creation. Inheriting traditional animation's protoplasmic characteristics, it uses models and data to redraw the power of images to “grow,” endowing digital animation with ever-changing color.

Although AIGC models increasingly display powerful capacities for self-growth and functions of subjectivity, an epochal gulf still separates assisted creation from autonomous generation, data models from organic life, and artificial intelligence from new intelligent entities. The investigation of digital imagery's plasmaticness reveals, at an ontological level, the ancestral connections between AIGC imagery, principally animation, and traditional animation. It also opens unlimited possibilities for future AIGC creation in the humanities.

Notes and References

References are numbered as in the original. Quotations in the main text are translated from the Chinese article; they have not been checked against the cited editions. Bibliographic details and factual claims are retained as printed, including apparent inconsistencies. The original's duplicate English abstract is represented by the translated abstract at the beginning.

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